CivicMitra (CivicOrbit)

Assignment 3 - Updated Prototype

Deliverables 1 to 5

1. Design & Implementation Report

Design Changes

- Fraud detection fixed - proof images are checked against the challenge's proof instructions before points are given
- Date feature added - current date shown on the landing page
- Website updated - landing page, verification result screens

Module-wise Implementation Details

Fraud Detection (Proof Verification)

- User uploads a proof photo for a challenge
- AI verification runs first, using the challenge title and proof instructions
- Result returned: verified, score, reason
- Not Approved: reason is shown and the user can Try Again
- Approved: points are awarded and the post appears in the feed with a Verified badge

AI Screen Scanner

- Options: Screen Grab, Snap Photo, Upload Proof
- Image is matched to an active challenge; verified only if matched and confidence is 0.70 or more
- Anti-cheat: image is hashed to prevent duplicate points
- Limits: 15 scans per day, 30-second cooldown

Date Feature

- Current date shown in a badge above the headline on the landing page
- Format: weekday, day, month, year
- Refreshes automatically every hour

2. Test Case Sheet

Test ID	Module	Input	Expected Output	Actual Output	Status
TC-01	Date Feature	Open landing page	Current date shown in badge	Badge shows Thursday, 1 October 2026	Pass
TC-02	Fraud Detection	Bucket Bath Challenge: upload photo of a glass of water	Verification not approved with reason	Verification Not Approved - image shows a glass of water being poured, not a bucket setup	Pass
TC-03	Fraud Detection	Rejected verification	Try Again button shown	Try Again button displayed	Pass
TC-04	Fraud Detection	Bucket Bath Challenge: upload photo of bucket setup	Proof verified, points awarded	Proof Verified - +15 points earned and streak kept	Pass
TC-05	Community Feed	Open feed after verified challenge	Verified post shown	Post #bucketbath shown with Verified badge	Pass

Test ID	Module	Input	Expected Output	Actual Output	Status
TC-06	AI Screen Scanner	Open AI Scanner	Screen Grab, Snap Photo, Upload Proof options and scan count shown	All options shown; 15 of 15 scans remaining today	Pass

3. Issue / Bug Log

Issue ID	Description	Severity	Root Cause	Fix	Status	Date
BUG-01	Fraud detection: irrelevant proof images could be accepted	High	Proof image not strictly checked against the challenge proof instructions	AI verification runs first and rejects non-matching images with a reason	Fixed	01-Oct-2026
BUG-02	No clear message after a rejected proof	Medium	No failure screen in the verification dialog	Added Verification Not Approved screen with reason and Try Again button	Fixed	01-Oct-2026
ENH-01	Date not shown on landing page	Low	Feature not implemented	Added date badge above the headline	Done	01-Oct-2026

4. Updated Source Code

- **GitHub repository link:** https://github.com/Project-Based-Learning-Project/CivicMitra
- **Tagged commit:** [Add tag, e.g. v1.1]
- Version notes: v1.1 - fraud detection fixed, date feature added, website updated

5. Updated Screenshots

5.1 Landing Page with Date Feature

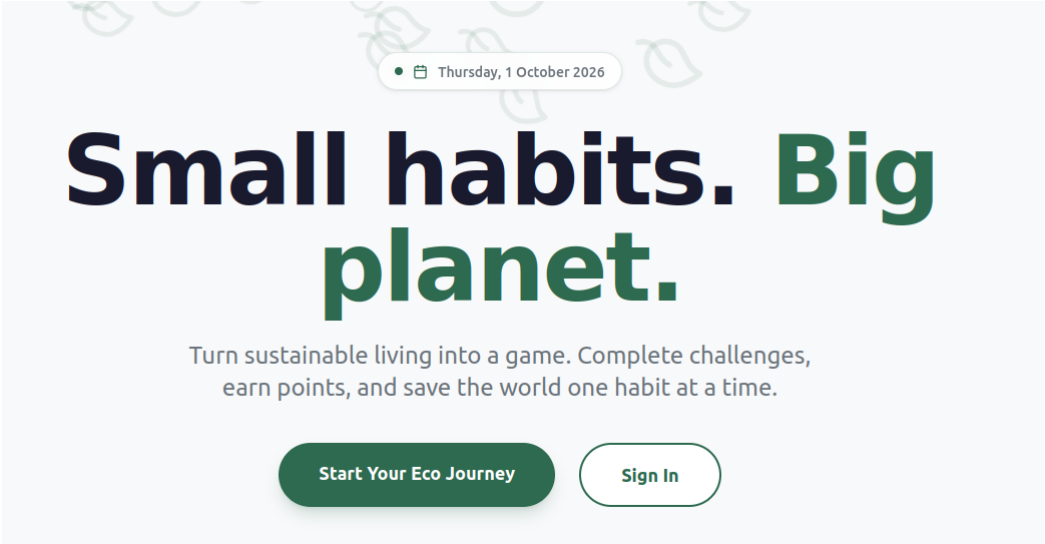


Figure 1: Date badge shown above the headline

5.2 Fraud Detection - Verification Not Approved

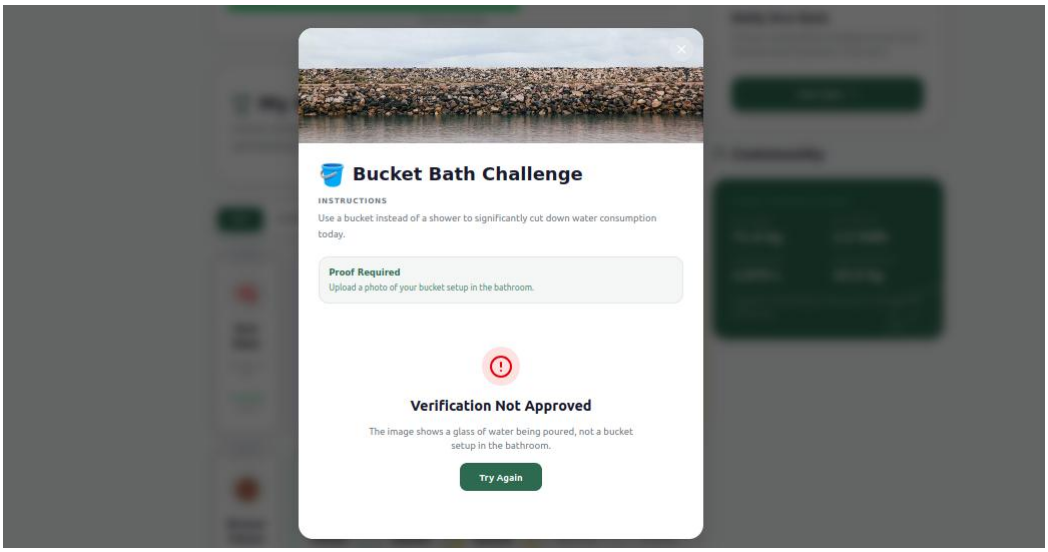


Figure 2: Wrong image rejected with a reason and Try Again button

5.3 Proof Verified

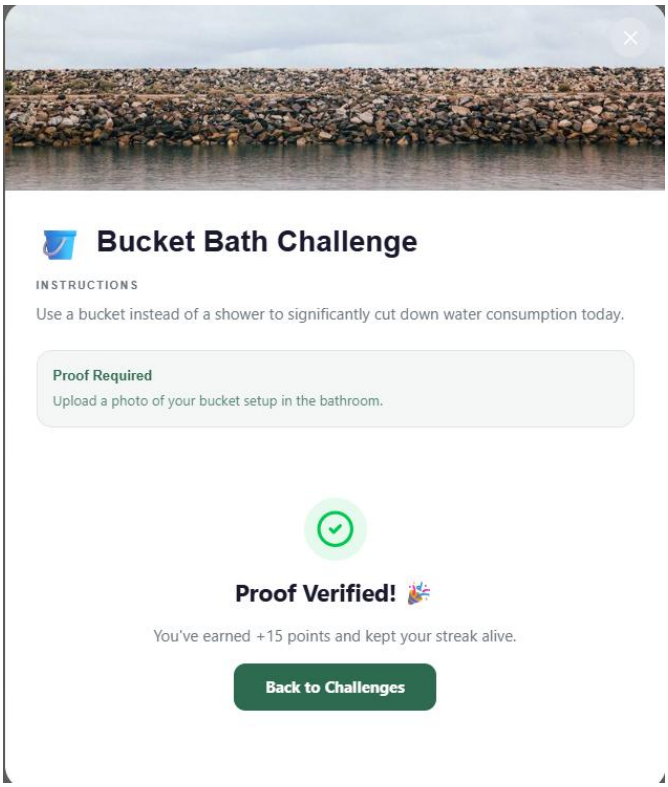


Figure 3: Correct proof approved, +15 points earned

5.4 AI Screen Scanner



Figure 4: Scanner with anti-cheat guarantee and scan limit

5.5 Community Feed

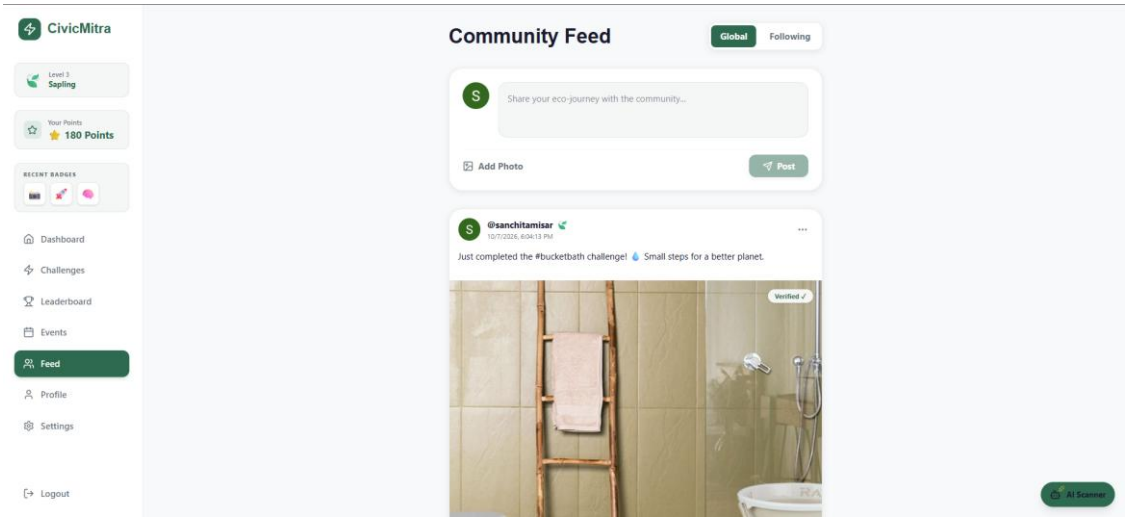


Figure 5: Verified challenge post in the feed